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Development of a standard for transient measurement of junction-to-case thermal resistance

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ABSTRACT

The paper summarizes the development of a standard method to measure the thermal resistance "junction-to-case" θ_{JC} of semiconductor devices with heat flow through a single path. Power switches or amplifiers are typical examples. θ_{JC} is a key performance metric to decide whether a device can be used in thermally critical applications. Hence an accurate and reproducible method to measure θ_{JC} is required. This is not easy, especially for low θ_{JC}, which is reflected by the fact that no JEDEC industry standard existed then to measure θ_{JC}. During the last 4 years we have evaluated approaches and developed a new method called Transient Dual Interface (TDI) method. It uses two measurements of the thermal impedance Zth or more specific $Z_{\theta JC}(t)$ of the device with different cooling conditions at the interface of device case and a heat sink. To evaluate these measurements two methods are applied. Method 1 determines θ_{JC} directly from the separation of Zth-curves. θ_{JC} is the thermal impedance $Z_{\theta JC}(t_s)$ at the time t_s where the two $Z_{\theta JC}(t)$-curves separate. Method 2 first calculates cumulative structure functions and uses their separation point to determine θ_{JC}. Both data evaluation methods complement each other, because method 1 is most accurate for low θ_{JC} in the range of 1 K/W or below, while method 2 is more accurate for higher θ_{JC} > 1 K/W. The TDI method allows to measure θ_{JC} with higher accuracy and better reproducibility than the steady state method used in industrial practice up to now. The TDI method was published as JEDEC standard JESD51-14 in November 2010. Problems of the traditional steady state measurement and main steps of the development of the TDI method are discussed.

1. Importance of thermal resistance "junction-to-case" θ_{JC} in power packaging

Trends in power electronics are as always in microelectronics towards more performance per cost, size and weight. The latter becomes more important as power supplies, chargers, regulators etc. go in mobile and automotive applications. Two of the key performance figures of a MOSFET power switch are "$R_{DS(\mathrm{on})}$" and "Rth_{JC}", shorter abbreviated as "θ_{JC}". $R_{DS(\mathrm{on})}$ is the electrical resistance R from Source to Drain connection in the transistor's "on" state which determines the power loss P_H converted to heat, while the "thermal resistance junction-to-case" Rth_{JC} or θ_{JC} indicates the technically best possible cooling performance at the device case to remove this heating power P_H and keep the device below the maximum allowed operating temperature $T_{\max}$.

For illustration let's assume a device with specified $T_{\max}$ = 150 °C in an application to conduct I = 50 A and in an ambient which can have 100 °C temperature, e.g. a motor. A device with $R_{DS(\mathrm{on})}$ = 20 mΩ converts $P_H = R_{DS(\mathrm{on})}{}^*I^2$ = 50 W to heat. With θ_{JC} = 1 K/W thermal resistance and ideal cooling at the device case we get a temperature increase of ΔT = 50 °C, which just meets $T_{\max}$. For θ_{JC} = 0.8 K/W we get ΔT = 40 °C, leaving a theoretical safety margin of 10 °C, and for θ_{JC} = 0.5 K/W we get ΔT = 25 °C, which might give the option to raise the allowed application temperature of the device. A reliable way to measure θ_{JC} could guarantee the thermal design targets to be fulfilled. However can we even distinguish the θ_{JC} cases discussed above in measurements? The measurement of small electrical resistances is no problem, while measurement of small thermal resistances is a big problem and even more challenging for small devices. Problems with the measurement of the case temperature T_{case} using thermocouples in the traditional steady state θ_{JC} – measurements are discussed here. Usually the thermal performance of microelectronic devices is measured according to procedures defined in the JEDEC JESD51 series of thermal standards. Despite of its high importance for power devices, no such standard existed until recent release of thermal standard JESD51-14 in November 2010 [1]. This paper summarizes motivation, steps, problems and solutions on the way to define this standard. Reference to more detailed publications to single topics is given.

The paper is organized as follows: It starts with the shortcomings of the traditional thermocouple test method, introduces transient measurement approaches, touches the main problems to be

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solved, and concludes with the standard and its requirements as a summary of the development efforts.

2. Traditional steady state measurement of θ_{JC}

The definition of θ_{JC} according to JESD51-1 [2] is: Thermal resistance, junction-to-case: the thermal resistance from the operating portion of a semiconductor device to outside surface of the package (case) closest to the chip mounting area when that same surface is properly heat sunk so as to minimize temperature variation across that surface.

2.1. Thermal contact at case to heat sink interface

The traditional measurement procedure roughly follows MIL Std. 833 [3]. As shown in Fig. 1 one package surface is pressed against a heat sink, usually a water cooled copper block. A layer of oil or thermal grease is applied at the interface to reduce the thermal contact resistance. The contact resistance is significant and might be the biggest thermal resistance in the flow path from the junction through chip, die attach, Cu die paddle, thermal grease and Cu-layer to flowing water, which is the final heat sink at constant reference temperature. The thermal contact resistance depends on surface roughness, interface layer thickness, pressure, potential dust particles and their variations. That is one problem for accurate measurement.

Power is applied to the device and after thermal equilibrium is reached, the dissipated (electrical) power P_H, junction temperature T_J and case temperature T_C are measured to calculate the steady state θ_{JC} as

$$\theta_{JC} = \frac{T_J - T_C}{P_H} \quad (1)$$

Looking closer, as it was done in our evaluations, this metric is less well defined as it seems to be in [4].

2.2. Thermocouple measurement of T-Case

T_J is measured electrically by means of a previously calibrated temperature sensitive parameter at the chip surface, usually the forward voltage of a diode. Using individual device calibration electrical measurement of T_J as well as of power P_H can be quite accurate. The problem is T_C, which usually is measured by a thermocouple bead contacting the cooled package case surface through a small hole in the copper block of the heat sink. This has several sources of potential error.

Ideally in order to have the thermocouple bead at the case temperature to be measured, it should have perfect thermal contact to the case and no heat losses to the surrounding heat sink, i.e. no contact in the first place. Best practice is to glue it with silicone rubber in a drill hole of 1–2 mm diameter with the thermocouple bead extruding just some tenth of millimeters above the heat sink surface. When the device is pressed against it, the thermocouple makes a point contact and moves in the flexible silicone, which acts like a spring and insulates to the heat sink, at least electrically. Perfect thermal insulation is simply not possible, because the contacting thermal grease should have good thermal conductivity and the surrounding heat sink material is well cooled and close by. Although silicone rubber has low thermal conductivity, the thermal insulation of the thermocouple wire by a thin layer is limited. The overall result is, that the thermocouple bead is not really at case temperature, but below between case and heat-sink temperature. The effect is investigated by simulations using the Finite Element (FE) model shown in Fig. 2. Thermal conductivity data of materials are listed in Table 1.

In the drill hole, half of the TC bead height is assumed to be filled with thermal grease. In all cases $P_H = 10$ W.

The result in Fig. 3 shows, that we already have a temperature gradient of 2.3 °C from top to bottom of the thermocouple bead. Which temperature will it measure? As in all electrical circuits it will output the voltage drop along the shortest electrical path between its thermal contacts at different temperature, i.e. the voltage drop for $T_{\text{TC-bot}}$. That leads to the results listed in Table 2 for different contact conditions of thermocouple and device case:

Even more error will be caused by a short circuit due to missing insulation of the two wires below the thermocouple bead, because it would return the temperature at the point of the short circuit instead of T_C.

2.3. Thermocouple point contact

Besides the thermal contact problem there is the potential difficulty of measuring temperature only in one point, assuming homogenous temperate over the surface. However, since the temperature distribution on the case surface is never truly uniform, the measured case temperature T_C always depends on the exact position of the T_C measurement. An example with hot spot is given in Fig. 4. In this case the deviation of θ_{JC} using the different two points to measure T_C is 12.5%.

2.4. Drill hole for thermocouple

The drill hole in the heat sink surface ideally should not disturb the temperature distribution at the case to heat sink interface. For a large power device with a massive metal pad of e.g. 2 mm thickness, a 1.5 mm drill hole in the heat sink is acceptable, but for QFP or QFN packages with exposed die pads sized down to $3 \times 3 \times 0.2$ mm^3 or even less it has a significant effect and the result will not reflect the thermal performance, when this device is

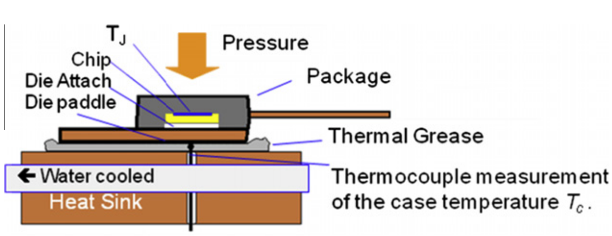

Fig. 1. Traditional steady state θ_{JC} thermal resistance measurement shown for a TO style package.

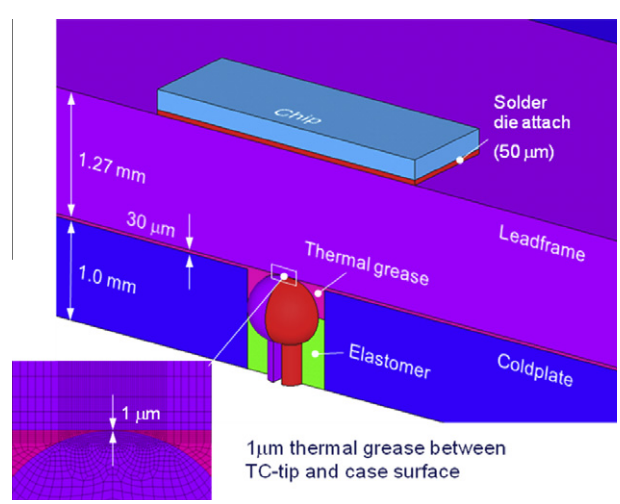

Fig. 2. Finite Element (FE) model for thermocouple evaluations.

Table 1
Thermal conductivity k used in FE models.

Material	k (W/(m K))
Chip	148
Solder die attach	53
Glue die attach	1
Cu die pad	350
Thermal grease	1.2
Elastomer (silicone rubber)	0.15
Chromel (thermocouple)	19.2
Alumel (thermocouple)	29.7

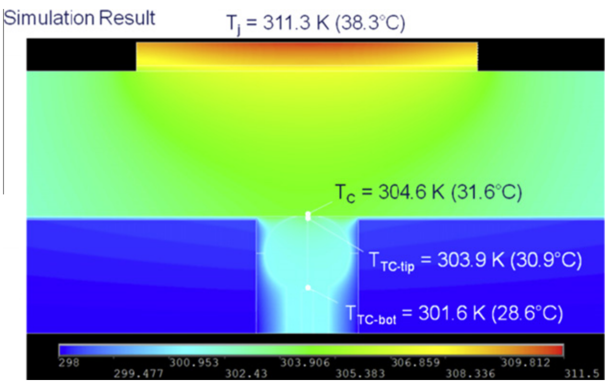

Fig. 3. Simulation result for thermocouple evaluations Note the temperature gradient in the thermocouple bead.

Table 2
FE simulation results of case temperature T_C and $T_{ThCouple}$ assumed to be T_C for different contact cases: (a) 1 μm thermal grease between ThermoCouple/case, (b) Direct metal point contact at thermocouple top, (c) ThCouple bead fully insulated to HS by silicone, (d) ThCouple bead has side contact to heat sink.

Case:	(a)	(b)	(c)	(d)
T_J	38.3	38.3	38.3	38.3
T_C	31.6	31.6	31.7	31.6
$T_{ThCouple}$	28.6	28.6	31.1	27.9
θ_{JC}–true	0.67	0.67	0.66	0.67
$\theta_{JC\text{-}ThCouple}$	0.97	0.97	0.92	1.04
θ_{JC}–Error (%)	**45**	**45**	**24**	**55**

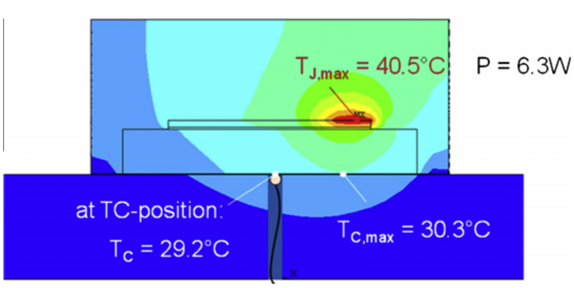

Fig. 4. Simulation result to illustrate the effect of temperature distributions on T_C measurement.

well cooled, e.g. soldered to a thermally high conductive PCB. An accurate performance metric is the main target. To be clear the traditional steady state θ_{JC} measurements can have high reproducibility in good test setups with positioning frames for device mounting and high pressure to assure good thermal contact. However this pressure will also close package internal delaminations, which increase θ_{JC} in a real application.

3. Transient Dual Interface Method (TDIM)

To overcome these problems a method was needed, which could characterize the package internal heat flow path from junction to case without measurement of the case temperature T_C by means of a thermocouple. Transient so called thermal impedance measurements are suitable and common practice for power devices. The results are part of data sheets to characterize pulse load thermal resistance, which is an important feature of power devices for short overloads and switching applications. Transient methods have been used to test internal thermal properties, e.g. die attach degradation, and have also been proposed for θ_{JC} measurement e.g. by Siegal [5]. Szabo et al. presented a method to identify θ_{JC} comparing the structure functions obtained by two transient measurements with different interface layers between package and heat sink [6,7]. The principle should work as all thermal resistances and capacities along a 1-dimensional heat flow path should be visible in structure functions [8,9]. An evaluation [10] revealed that the mathematical procedures to extract the structure functions from the impedance curve are error prone and noise sensitive especially at short times, corresponding to low thermal resistance. The basic idea remains, to separate the package internal thermal resistance from junction-to-case surface in contact to a heat sink, from the external contact and heat sink resistances. How to do that the best way? That is the main problem. The procedure has to be fixed in a standard to assure comparability of results. Main problems and solutions will be discussed. The method was called Transient Dual Interface (TDI) method.

3.1. TDI measurement principle

No JEDEC standard for transient thermal measurements existed. Therefore everything needed had to be defined.

Thermal impedance definition: The thermal impedance or Zth-function $Z_{\theta JC}(t)$ of a semiconductor device which is heated with constant power P_H starting at time $t = 0$ while its case surface is properly heat-sunk shall be defined as

$$Z_{\theta JC}(t) = \frac{T_J(t) - T_J(t=0)}{P_H} \tag{2}$$

i.e. the thermal impedance equals the time-dependent change of the junction temperature $T_J(t)$ divided by the heating power P_H. If the cooling condition at the package case is changed, this should have no influence on the thermal impedance until the temperature starts to increase at the package case where the contact to the heat sink is present. However a measurement with different contact resistance changes the total thermal resistance at steady-state and therefore separates the impedance curves of different measurements. Separation starts from the point where the external contact resistance contributes, which can be identified as the package case interface.

Two thermal impedance measurements of the same device with and without thermal grease at the package case to heat sink interface are done to identify the case surface in transient measurements. The principle is shown in Fig. 5. When plotted as a diagram, Zth-curves are shown in logarithmic time scale.

The higher interface resistance in case (I) due to the microscopic surface roughness between package and cold-plate ensures a clear separation of the Zth-curves at some point of time t_S. The value $Z_{\theta JC}(t_S)$ at that time shall be referred to herein as "transient θ_{JC}", since the separation of the two curves marks the transition of the heat flow from the package into the thermal interface layer.

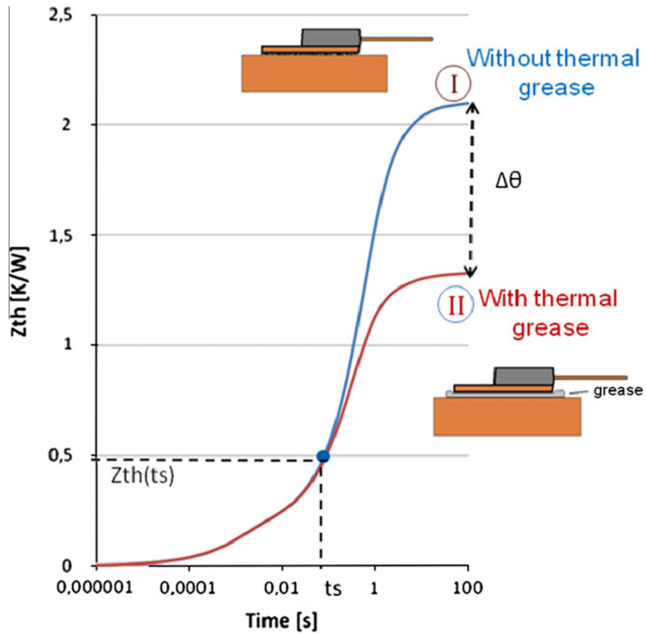

Fig. 5. Zth curves for a package measured with and without thermal grease at the interface between case and heat sink.

3.2. Offset correction

Switching the heating power on or off at the beginning of the Zth-measurement inevitably causes electrical disturbances of the signal, thus rendering the measurement useless for short times. The signal has to be discarded for all points of time t smaller than a certain cut-off time t_{cut}. In order to reconstruct the non-negligible temperature rise $\Delta T(t_{cut})$ in the time range before t_{cut} one can exploit the almost linear relationship between temperature increase and the square-root of time for short times t (Fig. 6).

For a "semi-infinite" plate (i.e. a plate with infinite surface area – ensuring 1-dimensional heat flow perpendicular to the surface – and infinite thickness) of homogeneous material, for the surface, which is heated with constant power density P/A, it can be shown (see e.g. [11]) that the surface temperature rises linearly with the square root of heating time:

$$\Delta T(t) = \frac{P}{A} k_{therm} \sqrt{t} \tag{3}$$

with

$$k_{therm} = \frac{2}{\sqrt{\pi c \rho \lambda}} \tag{4}$$

c, ρ, and λ being the specific heat, density, and thermal conductivity of the plate material, respectively. For short times during which the heat propagation inside the silicon chip is approximately 1-dimensional and undisturbed by boundary effects from the chip bottom surface, the "semi-infinite" plate model can be applied to the heating of the chip surface. This is confirmed by measurements, as shown in Fig. 6. The initial temperature rise $\Delta T(t_{cut})$ can easily be found extrapolating the $\Delta T(t)$ vs. t graph to $t = 0$. As a side benefit an estimate of the active chip area A can be obtained by substituting the known slope $m = \Delta T(t)/\sqrt{t}$ of the linear fit into Eq. (3), which results in

$$A = \frac{P}{m} k_{therm} \tag{5}$$

A realistic chip area A calculated this way can serve as cross-check for a reasonable offset correction.

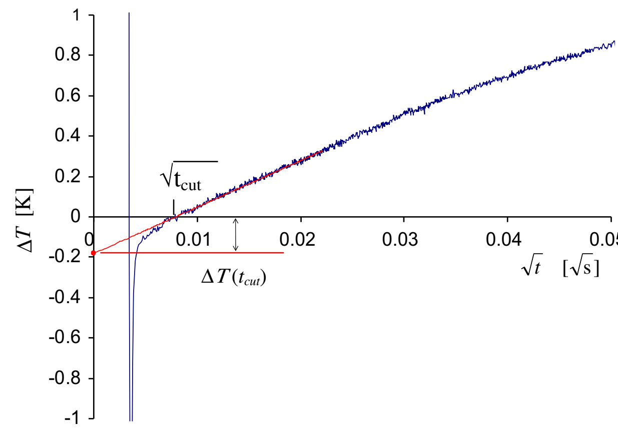

Fig. 6. Measured temperature response of a MOSFET chip to a constant power pulse vs. t. For short times t the chip temperature rises approximately linear with the square-root of pulse-duration t.

3.3. Evaluation of the TDI measurement

A closer look at the Zth curves in Fig. 5 shows that there is no well defined point of separation. In fact, the gap between the curves widens gradually over some range of time. In order to determine the point of separation of the two Zth curves it is better to examine the derivatives da/dz of the Zth-curves instead of the Zth-curves [10,12]. $a(z)$ denotes the Zth-value as a function of logarithmic time $z = \ln(t)$, i.e. $a(z) = Z\text{th}(t)$. Thus da/dz is just the slope of the Zth-curve in the usual log-linear representation. Of course the curves in Fig. 5 will be shifted by the offset correction, which has an impact on the separation point.

Using derivatives eliminates potential errors from different offsets of the two Zth curves. Going one step further in data evaluation by plotting the absolute difference $\Delta(da/dz)$ as shown in curve (c) of Fig. 7, makes it much easier to identify the separation point, because it is marked by a steep increase of that graph. Nevertheless separation is still not exactly defined. A closer look shows

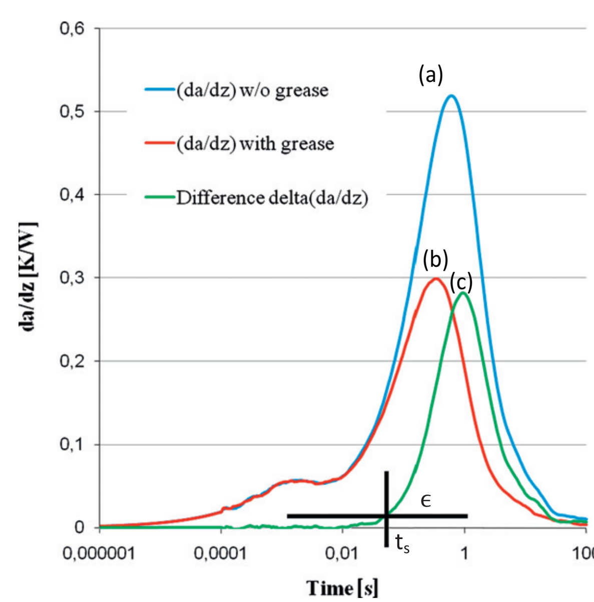

Fig. 7. Derivatives (da/dz) and the difference $\Delta(da/dz)$ of the Zth-curves from Fig. 5. A threshold $\Delta(da/dz) = \varepsilon$ has to be chosen to define separation time t_s.

fluctuations due to inevitable noise in the measurements and steady increase. A threshold distance "ε" has to be chosen to clearly define the time t_s for separation. Taking $Zth(t_s)$ from the impedance curve would then give the "transient θ_{JC}". A more convenient direct read-out is possible by plotting $\Delta(da/dz)$ not over time, but over the corresponding impedance Zth as shown in Fig. 8. The "transient θ_{JC}" obtained that way is a function of the maximum distance "ε" for which the derivatives da/dz are regarded as equal.

3.4. How to choose the limit ε to determine θ_{JC}

Method 1: Point of separation of the $Z_{\theta JC}$ curves: The question is how to choose "ε" such that the transient θ_{JC} defined by this "ε" is equal to the steady-state θ_{JC} which we want to measure. The search for "ε" starts with the observation that the $|\Delta(da/dz)|$ vs. Zth curve shown in Fig. 8 is influenced by the maximum distance $\Delta\theta$ of the two Zth-curves in steady-state (Fig. 5). $\Delta\theta$ corresponds to the increase of the thermal resistance of the heat-flow path due to the lack of thermal grease in one measurement. It depends on many parameters such as contact area, thickness and material properties of the thermal interface layer, surface roughness of package and heat sink, and the pressure used to clamp the device to the heat sink. The larger $\Delta\theta$, the steeper the slope of $\Delta(da/dz)$.

In order to minimize the influence of this randomly distributed variable, $\Delta(da/dz)$ should be normalized by $\Delta\theta$. We therefore defined the (dimensionless) function

$$\delta(Z_{th}) \equiv \frac{\Delta(da/dz)}{\Delta\theta} \tag{6}$$

and search for a value ε such that the "transient θ_{JC}" obtained with this ε is exactly equal to the steady-state θ_{JC} which we want to measure. In order to prevent that the result is falsified by random noise of the δ-curve, it shall be replaced by an appropriate fit function, e.g. an exponential fit curve

$$\delta = \alpha \cdot \exp(\beta \cdot Z_{\theta JC}) \tag{7}$$

with parameters α and β (Fig. 8) optimized for good fitting in the region around θ_{JC}.

Since for real semiconductor devices the actual steady-state θ_{JC} is a priori unknown, we have to rely on finite element models for which we can compute the θ_{JC}. Using a simplified FE-model of a power semiconductor device on a cold-plate, we computed both the steady-state θ_{JC} using Eq. (1) and the Zth-curves of the device on the heat sink with and without thermal grease layer. That way it was possible to determine the "correct" ε for which "transient θ_{JC}" equals "steady-state θ_{JC}". Details of these simulations are published in [13]. Simulations were performed for different chip sizes, two different die-pad thicknesses (0.25 mm and 1.27 mm) and for devices with glue and solder die attach. Results are shown in Fig. 9.

Observations:

(a) ε increases approx. linearly with θ_{JC} in all cases,
(b) die pad thickness has a small effect (solder die attach),
(c) devices with glue die attach have 5–10 times higher ε and also θ_{JC} compared to those with solder die attach.

Neglecting the small influences of die pad thickness, the following general trend-line could be fitted for power devices with high conductive die attach like solder:

$$\varepsilon = 0.0045\ \mathrm{W/K} \cdot \theta_{JC} + 0.003 \tag{8}$$

With this results method 1 to determine θ_{JC} of devices with thermally high conductive die attach, e.g. solder could be defined [1]:

Method 1: Determination of θ_{JC} based on the point of separation of the $Z_{\theta JC}$ curves: The intersection point of fit function (7) and the ε-curve (8) is the junction-to-case thermal resistance θ_{JC}.

The evaluation is shown in Fig. 8. Minor influence factors on ε, e.g. as shown the die pad thickness, put a limit to the achievable accuracy and reproducibility of the measurement [14], but for the boundary conditions specified in the JEDEC standard [1], Eq. (8) approximates the separation distance ε closely enough. If it turns out in practice that dedicated ε-curves (8) are needed for other package constructions with high conductive die attach, they can be added later.

However for devices with low conductive glue die attach layer in the heat flow path, this method 1 would give wrong results. As an example for θ_{JC} = 1 K/W the separation distance ε has to be about ε = 0.03 (Fig. 9), which is four times the value given by (8). That means the separation between impedance curves must already be large at the point corresponding to the steady state θ_{JC}, or in other words a visible separation of the impedance curves occurs too early to be an indicator for θ_{JC}, and if used the θ_{JC} value would be much too low. Obviously there is already a temperature increase and thermal flux through the external case-to-cold plate interface at separation time t_s, before the package internal flux distribution is fully developed as in the steady state. The explanation is that an internal heat-flow barrier delays the increase of the heat-flux inside the device, i.e. the difference between the heat-fluxes at

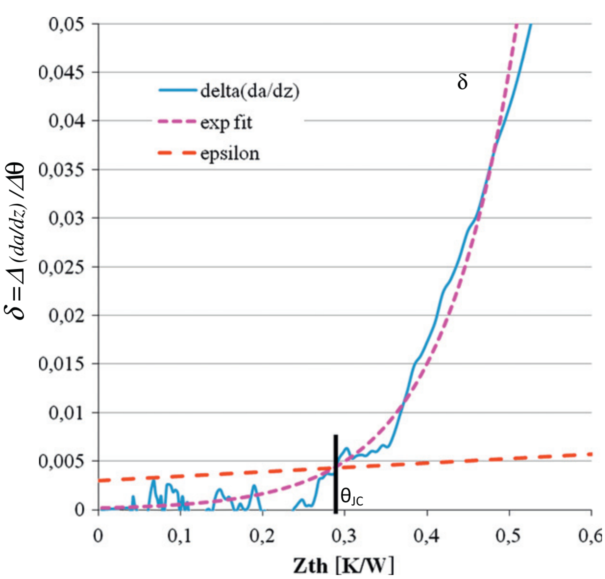

Fig. 8. Normalized difference $\delta = \Delta(da/dz)/\Delta\theta$ of the derivatives plotted vs. Zth. The θ_{JC} is the point of intersection of ε and δ.

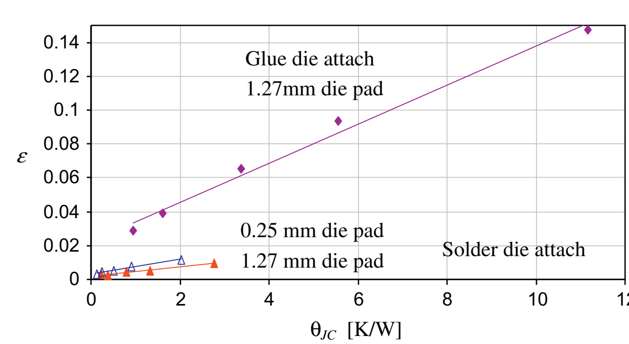

Fig. 9. ε vs. steady-state θ_{JC} for glue die attach and solder die attach devices as computed by FE simulation for different chip sizes.

time t_s and those in steady state becomes bigger. Therefore we have to treat devices with thermally high conductive solder die attach and devices with low conductive glue die attach differently.

3.5. Alternative evaluation method 2 for glue die attach devices

For devices with thermally low conductive die attach we therefore cannot evaluate the point of separation directly from Zth-curves. Instead we get back to the original suggestion of Szabo et al. [6,7] and evaluate the separation point of the structure functions. This method should in principle be equally applicable to glue and solder die attach devices; however the limited resolution of the structure function makes an accurate evaluation of measurements of low θ_{JC} devices quite difficult [13,14]. On the other hand the error associated with the evaluation of the structure function is almost independent of the actual θ_{JC} value. Compared to the larger θ_{JC} value of glue die attach devices the error of the structure function method is in most cases acceptable whereas it might not be acceptable for solder die attach devices with a very low θ_{JC}.

Other than suggested in [6] we recommend evaluating the cumulative structure function instead of the differential structure function. Inevitable numerical disturbances of the structure function due to noise in the measured Zth-curve are further enhanced in the differential structure function. Fig. 10 shows the structure functions of a glue die attach device measured with and without thermal grease interface.

The difference of these two curves is plotted in Fig. 11. They separate at $\theta_{JC} \approx 4.2$ K/W, which is well detectable. According to this method 2, the junction-to-case thermal resistance θ_{JC} is the point where the difference $\Delta C_{\theta\Sigma}$ clearly starts to rise. A thorough discussion of the error associated with this evaluation can be found e.g. in [14]. We estimate that the error is less than ±20%.

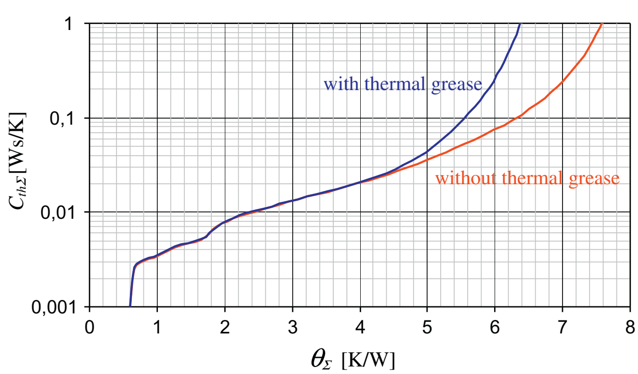

Fig. 10. Cumulative structure functions of a glue die attach device (measurement). The cumulative heat capacitance $C_{th\Sigma}$ along a 1-dimensional heat flow path is plotted over the corresponding cumulative resistance θ_Σ.

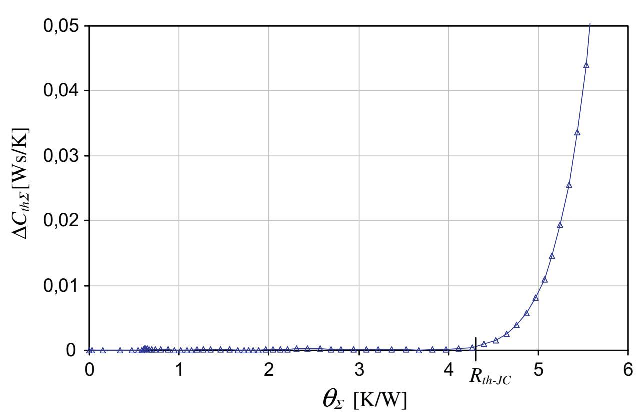

Fig. 11. Difference $\Delta C_{th\Sigma}$ of the cumulative structure functions (measurement) in Fig 10.

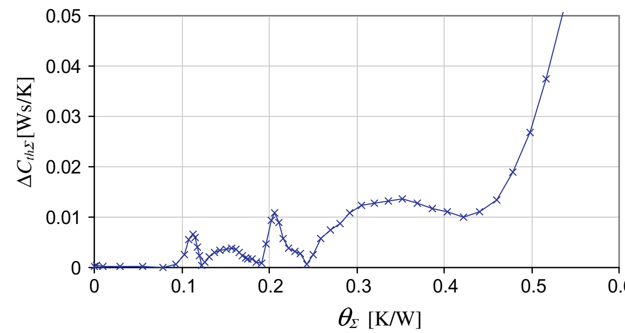

Fig. 12. Difference of the cumulative structure functions of a low θ_{JC} device (measurement).

Fig. 12, on the other hand, shows the difference of the two structure functions for a solder die attach device with a very low $\theta_{JC} \approx 0.25$ K/W. Due to numerical effects of the deconvolution algorithm used to calculate structure functions from Zth-curves, the difference $\Delta C_{th\Sigma}$ is seldom a smooth curve for low θ_{JC} packages and the uncertainty of the position of the separation point is by far too large compared to the small θ_{JC} value itself. Deconvolution should be avoided here. On the other hand method 1 works well for low θ_{JC} packages. Methods 1 and 2 to measure θ_{JC} complement each other. Therefore they both have become part of the standard [1] to achieve the best possible accuracy of θ_{JC} measurements.

4. Methods and requirements in the standard

Conclusions of investigations over several years are summarized in the methods and requirements defined in JEDEC standard JESD51-14 [1].

Two thermal impedance $Z_{\theta JC}(t)$ measurements with different contact resistance for cooling the heat sunk package case surface are done to identify the package case interface in transient measurements. The cumulative thermal resistance at the separation point of these two measurements is defined as $R_{\theta JC}$ (θ_{JC}).

The two TDI measurements are fast and straight forward. Recording the cooling curves instead of heating is recommended because the power P_H is kept constant. Further a minimum sampling rate is required for later data evaluation, e.g. calculation of derivatives.

Just the TDI data evaluation needs some mathematical procedures which are not always used or available. Therefore a software package called "TDIM Master" is provided and can be downloaded from the JEDEC website along with the standard. It is intended as an example how to do it. Other software might be used as long as the results agree with benchmark cases given in the standard as Appendix D. The TDIM-Master software can read in $Z_{\theta JC}(t)$ measurement files in some formats, e.g. ASCII. All mathematical procedures discussed above are implemented, i.e. offset correction, calculation of derivatives, normalized differences, curve fitting, calculation of the cumulative structure functions and finally output of θ_{JC1} and θ_{JC2} for both evaluation methods. The higher value has to be reported as thermal resistance junction-to-case θ_{JC}.

Step by step evaluation procedures are described in the standard. Relevant background information, e.g. on cumulative structure function and RC-networks is added as Appendices A, B and C for understanding and as base for future standards using transient methods.

First round robin results from measurements of some MOSFET devices at different laboratories indicated, that the cooling effi-

Table 3
Reproducibility test example. θ_{JC} evaluation of (10 + 10) $Z_{\theta JC}$ measurements of one MOSFET device.

θ_{JC} [K/W]	Measurement with thermal grease									
	1	2	3	4	5	6	7	8	9	10
Measurement w/o grease										
1	1.19	1.09	1.07	1.10	1.16	1.15	1.13	1.15	1.19	1.16
2	1.24	1.17	1.14	1.16	1.21	1.21	1.19	1.21	1.23	1.21
3	1.23	1.14	1.12	1.14	1.20	1.20	1.17	1.19	1.22	1.20
4	1.23	1.14	1.12	1.14	1.19	1.19	1.17	1.19	1.21	1.20
5	1.24	1.16	1.13	1.15	1.20	1.20	1.18	1.20	1.22	1.20
6	1.26	1.20	1.17	1.18	1.23	1.23	1.21	1.23	1.24	1.23
7	1.27	1.22	1.19	1.20	1.24	1.25	1.23	1.25	1.25	1.24
8	1.27	1.21	1.17	1.20	1.24	1.24	1.22	1.25	1.25	1.24
9	1.26	1.20	1.17	1.18	1.23	1.23	1.21	1.24	1.25	1.23
10	1.27	1.20	1.18	1.19	1.23	1.24	1.22	1.24	1.25	1.23

Average *R*th-*JC*: 1.20 K/W.
Standard deviation: 0.04 K/W (3.5%).
Maximum deviation: 0.13 K/W (11%).

ciency of the heat sink used influences the results. To avoid this, the requirements here became part of the standard. To approximate ideal cooling, the heat sink should be very efficient, i.e. the mounting base shall consist of a copper block, the temperature of which is kept constant by a cooling fluid (usually water) passing through drilled holes in the block. The fluid should flow below the mounting base leaving maximum 2 mm Cu-material between drill hole for water cooling and mounting base surface in contact with the cooled package case surface.

A detailed discussion on reproducibility and accuracy as well as limitations of the TDI method is given in [14]. Due to the lack of reliable θ_{JC} reference values that is not easy. The accuracy of TDI results is estimated to be ±20% and probably better in most cases.

The reproducibility of TDI θ_{JC} measurements is usually much better. Table 3 shows the result of repeated measurements of one MOSFET device performed with the same measurement equipment. Both the dry measurement and the measurement with thermal grease interface have been repeated 10 times, thus allowing 10×10 pair-wise evaluations. The average θ_{JC} value of this matrix is 1.20 K/W with a maximum deviation of 0.13 K/W (10.9%) and a standard deviation of just 0.04 K/W (3.5%).

The TDI method relies only on the measurement of the junction temperature. It also does not need high pressure to assure good thermal contact to the heat sink. Moderate pressure in the range of 10 N/cm^2 is specified to keep it in place.

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